KNECK OFFICIAL

Agriculture — Paper 2

443/2 • ORIGINAL REVISION SET 01 • 2 hours • 80 marks

Name: _____ Class: _____ Date: _____

Independent KCSE-style practice. Not issued, endorsed or predicted by KNEC. These are original questions, not leaked or past official papers. Component numbering is checked against the KNEC 2026 timetable. A teacher should review these practice materials before formal classroom assessment.

CANDIDATE INSTRUCTIONS

Answer all Sections A and B and ANY TWO essays in Section C. Use local livestock examples and safe, welfare-conscious practice.

Use a separate answer booklet. Show working where appropriate. The marking guidance begins after the END OF QUESTIONS page; do not consult it until you finish. Practical and project components require the stated equipment and teacher supervision, not merely a printed worksheet.

SECTION A

30 marks.

1. Give three signs of a healthy animal. [3 marks]

2. State three reasons for selecting breeding stock. [3 marks]

3. Name three livestock parasites and classify each as internal/external. [3 marks]

4. State three components of a balanced livestock ration. [3 marks]

5. Give three advantages of artificial insemination. [3 marks]

6. State three uses of farm fences. [3 marks]

7. Give three maintenance practices for hand tools. [3 marks]

8. State three signs a cow may be on heat. [3 marks]

9. Give three ways to prevent milk contamination. [3 marks]

10. State three benefits of rotational grazing. [3 marks]

SECTION B

20 marks.

11. A dairy ration mixes 60kg feed A containing 10% crude protein with 40kg feed B containing 25%. (a) Calculate protein mass in each and percentage in mixture. (6) (b) State two limits of using crude protein alone to judge ration quality. (4) [10 marks]

12. A cow yields 12L/day for 280 days. Milk sells at KSh50/L; variable costs total KSh100,000. (a) Calculate revenue and gross margin. (4) (b) Explain three ways of improving herd productivity without compromising welfare. (6) [10 marks]

SECTION C

Choose two; 15 marks each.

13. Describe management of a calf from birth to weaning, including hygiene, feeding, identification and disease prevention. [15 marks]

14. Explain establishment and management of a productive pasture for dairy cattle. [15 marks]

15. Describe routine maintenance and safe use of a tractor and an animal-drawn implement; explain economic factors in choosing mechanisation. [15 marks]

END OF QUESTIONS

MARKING GUIDANCE

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Suggested answers and assessment guidance, not an official marking scheme. Accept equivalent correct reasoning, valid examples and properly justified alternative methods. Marks for an item must not exceed its printed allocation. For essays use relevant textual evidence and the rubric; no plot summary alone.

SECTION A

1. (3 marks) Alertness,normal appetite/temperature,normal movement,clean eyes/coat,three.

2. (3 marks) Productivity,health/resistance,adaptation,temperament,fertility,three.

3. (3 marks) Tick external,louse external,roundworm/tapeworm internal,three paired.

4. (3 marks) Energy,protein,minerals,vitamins,water/fibre with species context,three.

5. (3 marks) Access superior genetics,disease risk reduction with hygiene,cost/transport,records,three.

6. (3 marks) Security,stock control,paddocks,crop protection,boundaries,three.

7. (3 marks) Clean,dry/lubricate,sharpen safely,repair handles,store safely,three.

8. (3 marks) Standing to be mounted,restlessness,mucus discharge,mounting others,reduced feed,three.

9. (3 marks) Clean udder/hands/equipment,rapid cooling,healthy animals,covered storage,three.

10. (3 marks) Pasture recovery,better utilisation,parasite disruption,stock control,reduced selective overgrazing,three.

SECTION B

11. (10 marks) A6kg,B10kg,total16kg/100kg=16% (6). Digestibility/amino acid balance,energy/minerals,fibre,mycotoxins/species stage,two explained (4).

12. (10 marks) Milk3360L,revenue168,000,gross margin68,000 (4). Balanced feeding,preventive health,veterinary care,comfortable housing,good genetics/records,three explained (6).

SECTION C

13. (15 marks) Clear airways/dry/warm and navel hygiene3; early adequate colostrum3; clean milk/water/starter/forage as appropriate3; housing/records/identification3; health monitoring/vet plan/humane gradual weaning3. No invasive procedure should be attempted without trained supervision.

14. (15 marks) Site/species/soil testing3; seedbed/planting3; nutrients/water2; weeds/pests2; rotational grazing/stocking/recovery3; conservation/records2.

15. (15 marks) Tractor fluids,tyres,guards,service schedule/safe start5; implement inspection,hitching,adjustment,cleaning/storage and animal welfare5; farm size,cost,labour,timeliness,terrain,repair access five economic/context points5. Never operate machinery untrained.

REVISION CHECK

Mark your work, list three misconceptions, revise those topics and attempt the incorrect questions again without the guide. These topics are revision priorities, not a forecast of the real examination.

Paper/component reference: https://www.knec.ac.ke/wp-content/uploads/2026/02/2026-KCSE-Timetable.pdf